where, E_o and B_o are amplitude of oscillating electric and magnetic field, k is a wave number, ω is the angular frequency of the wave and $\hat{k}$ (unit vector, here it is called propagation vector) denotes the direction of propagation of electromagnetic wave.

Note that both electric field and magnetic field oscillate with a frequency (frequency of electromagnetic wave) which is equal to the frequency of the source (here, oscillating charge is the source for the production of electromagnetic waves). In free space or in vacuum, the ratio between E_o and B_o is equal to the speed of electromagnetic wave, which is equal to speed of light c.

$$c = \frac{E_\circ}{B_\circ}$$

In any medium, the ratio of E_o and B_o is equal to the speed of electromagnetic wave in that medium, mathematically, it can be written as

$$v = \frac{E_\circ}{B_\circ} < c$$

Further, the energy of electromagnetic waves comes from the energy of the oscillating charge.

EXAMPLE 5.3

Compute the speed of the electromagnetic wave in a medium if the amplitude of electric and magnetic fields are 3×10^4 N C^{-1} and 2×10^{-4} T, respectively.

Solution

The amplitude of the electric field, $E_o = 3 \times 10^4$ N C^{-1}

The amplitude of the magnetic field, $B_o = 2 \times 10^{-4}$ T. Therefore, speed of the electromagnetic wave in a medium is

$$v = \frac{3\times10^4}{2\times10^{-4}} = 1.5\times10^8\, m s^{-1}$$

5.2.3 Electromagnetic spectrum

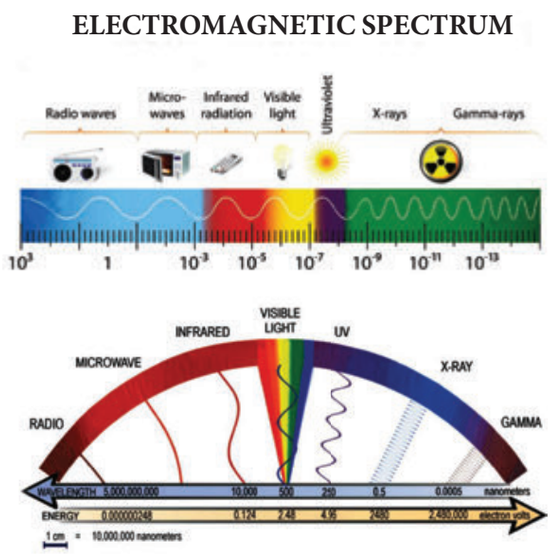


Figure 5.10 Electromagnetic spectrum -

Electromagnetic spectrum is an orderly distribution of electromagnetic waves in terms of wavelength or frequency as shown in Figure 5.10.

Radio waves

It is produced by oscillators in electric circuits. The wavelength range is 1×10^{-1} m to 1×10^4 m and frequency range is 3×10^9 Hz to 3×10^4 Hz. It obeys reflection and diffraction. It is used in radio and television communication systems and also in cellular phones to transmit voice communication in the ultra high frequency band.

Microwaves

It is produced by electromagnetic oscillators in electric circuits. The wavelength range is 1×10^{-3} m to 3×10^{-1} m and frequency range is 3×10^{11} Hz to 1×10^9 Hz. It obeys reflection and polarization. It is used in radar system for aircraft navigation, speed of the vehicle, microwave oven for cooking and very long distance wireless communication through satellites.